Electric Field of a Parallel-Plate Capacitor

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PHSX 333 — Project 2

1 Introduction

This project investigates the electric potential and electric field inside a grounded square box containing two parallel conducting plates held at opposite potentials $V = +1$ and $V = -1$. The plates are half as tall as the side of the box, centered vertically, and positioned so that one-third of the box lies to the left of the negative plate, one-third between the plates, and one-third to the right of the positive plate. The surrounding box walls are grounded ($V = 0$).

Inside the box, in the absence of free charges, the electric potential obeys Laplace's equation:

$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} = 0 \tag{1}$$

No analytic solution exists for this geometry, so the equation must be solved numerically. Two iterative relaxation methods are implemented and compared: the basic Jacobi method and Successive Over-Relaxation (SOR).

Qualitative Expectations

Between the two plates, the field lines are expected to run nearly horizontally from the positive plate to the negative plate, closely resembling the uniform field of an ideal parallel-plate capacitor. Outside the plate region, the field fans out into fringe fields that decay toward the grounded walls.

As the plates are brought closer together the potential difference remains fixed at $\Delta V = 2$, so the electric field magnitude between them, which scales as $|E| \approx \Delta V/d$, must increase as the separation d decreases. At the same time, fringe effects become a progressively smaller fraction of the total field, and the interior field becomes more uniform. In the limit $d \to 0$ the field between the plates diverges while the field everywhere else vanishes.

2 Numerical Methods

2.1 Finite-Difference Discretisation

The square domain is discretised onto an $N \times N$ grid with grid spacing $\Delta x = \Delta y = L/(N-1)$. Applying a second-order central finite difference to both spatial derivatives gives the standard five-point stencil:

$$V_{i,j} = \frac{1}{4}\big(V_{i+1,j} + V_{i-1,j} + V_{i,j+1} + V_{i,j-1}\big) \tag{2}$$

This is the update rule used by both solvers. Boundary conditions are enforced at every iteration: the four walls are held at $V = 0$, the left plate at $V = -1$, and the right plate at $V = +1$. Both solvers run until the convergence criterion

$$\max_{i,j} \big| V_{i,j}^{\text{new}} - V_{i,j} \big| < \varepsilon \tag{3}$$

is satisfied, where $\varepsilon = 10^{-4}$ throughout unless otherwise stated.

2.2 Jacobi Method

The Jacobi method updates every interior grid point simultaneously using the values from the previous iteration. A full copy of the grid is made at each step; the new values are computed from the old copy and then replace it. This is the simplest relaxation scheme but also the slowest to converge.

2.3 Successive Over-Relaxation (SOR)

SOR accelerates convergence by introducing a weighting parameter $\omega \in (1, 2)$:

$$V_{i,j}^{\text{new}} = (1-\omega)\, V_{i,j} + \omega \cdot \frac{1}{4}\big(V_{i+1,j} + V_{i-1,j} + V_{i,j+1} + V_{i,j-1}\big) \tag{4}$$

For $\omega = 1$ SOR reduces to the standard Gauss–Seidel method. For $\omega > 1$ the update overshoots the Gauss–Seidel value, which can dramatically reduce the number of iterations required. A red-black (checkerboard) ordering of the grid points is used so that each colour can be updated as a vectorised array operation while still propagating information in both directions within a single sweep, preserving the stability of the scheme for all $\omega \in (0, 2)$.

2.4 Electric Field Calculation

Once the converged potential grid is obtained, the electric field is computed via the relation $\mathbf{E} = -\nabla V$. Central finite differences are used at interior points:

$$E_x \approx -\frac{V_{i,j+1} - V_{i,j-1}}{2\,\Delta x}, \qquad E_y \approx -\frac{V_{i+1,j} - V_{i-1,j}}{2\,\Delta y} \tag{5}$$

implemented in practice using `numpy.gradient`. The result is displayed as a quiver plot in which each arrow is normalised to unit length and coloured by $|\mathbf{E}|$.

3 Results

3.1 Optimal SOR Weighting (Part c)

Using a grid of $N = 100$ points per side and $\varepsilon = 10^{-4}$, the number of iterations required for convergence was measured over a range of SOR weights. A coarse sweep from $\omega = 1.0$ to $\omega = 1.9$ in steps of 0.05 was followed by a fine sweep from $\omega = 1.80$ to $\omega = 1.92$ in steps of 0.01. The potential distributions for SOR at $\omega = 1.5$ are shown in Figures **??**–**??**, and the corresponding Jacobi solutions in Figures **??**–**??**. Both methods converge to the same potential distribution; they differ only in the number of iterations required.

SOR weight ω	Iterations to convergence
1.00	692
1.20	518
1.40	372
1.60	244
1.80	121
1.87	**68**
1.90	84
1.95	174

Table 1: Selected SOR iteration counts for $N = 100$, $\varepsilon = 10^{-4}$. The minimum occurs at $\omega = 1.87$.

The iteration count decreases steadily as ω increases from 1.0 to 1.87, then rises sharply as ω approaches 2, where the scheme becomes unstable. The optimal weight is $\omega^* = 1.87$, which requires only 68 iterations compared to 692 for the plain Gauss–Seidel case ($\omega = 1$) — a factor of ten reduction. The theoretical optimal for a rectangular domain is $\omega^* = 2/(1+\pi/N)$, which gives ≈ 1.94 for $N = 100$; the measured value of 1.87 is in reasonable agreement.

3.2 Convergence Scaling with Grid Size (Part d)

Using the optimal weight $\omega^* = 1.87$ for SOR and $\varepsilon = 10^{-4}$, both solvers were run for $N \in \{50, 100, 200, 400\}$. The results are plotted in Figure **??**. Power-law fits to the data

yield:

$$\text{Jacobi:} \quad \text{iterations} \propto N^{0.86} \tag{6}$$

$$\text{SOR:} \quad \text{iterations} \propto N^{1.26} \tag{7}$$

The theoretical predictions are N^2 for Jacobi and N^1 for optimal SOR. The measured exponents are lower than the theory for both methods, primarily because the small-N end of the range ($N = 50$) lies in a regime where the iteration count has not yet entered its asymptotic power-law behaviour, pulling the fit below the true large-N exponent. Nonetheless, the data clearly show that SOR scales considerably more favourably than Jacobi: at $N = 400$, Jacobi required 2414 iterations versus only 748 for SOR — a factor of ~ 3 at this grid size, growing with N.

3.3 Electric Field (Part e)

The converged potential for the default geometry (plate separation $d = L/3$) is shown as a greyscale density plot in Figure **??**, and the corresponding electric field quiver plot is shown in Figure **??**. Between the plates the arrows are dense, nearly horizontal, and of large magnitude, consistent with the expected uniform capacitor field. At the plate tips the field fans outward into fringe fields that decay toward the grounded walls, with the field pointing away from the positive plate and toward the negative plate in all regions.

Figures **??**–**??** show the electric field as the plate separation is reduced to $L/5$, $L/7$, $L/9$, and $L/11$. As predicted in the introduction, the inter-plate field grows stronger and more uniform with decreasing separation, while fringe effects become increasingly confined to the plate edges. This behaviour is consistent with the parallel-plate approximation $|E| \approx \Delta V/d$, which diverges as $d \to 0$.

4 Conclusion

This project successfully computed the electric potential and field inside a grounded square box containing two oppositely charged parallel plates using two iterative relaxation methods: the Jacobi method and Successive Over-Relaxation (SOR).

For a 100×100 grid, the optimal SOR weighting parameter was found to be $\omega^* = 1.87$, reducing the number of iterations by a factor of ten compared to the unaccelerated Gauss–Seidel method. A convergence scaling study over $N \in \{50, 100, 200, 400\}$ confirmed that SOR converges significantly faster than Jacobi at all grid sizes tested, with SOR requiring roughly three times fewer iterations at $N = 400$.

The computed electric field matches qualitative expectations throughout: a nearly uniform, horizontally directed field between the plates and decaying fringe fields outside them. As the plates were brought closer together, the inter-plate field strength increased and became more uniform, in agreement with the parallel-plate approximation $|E| \approx \Delta V/d$.

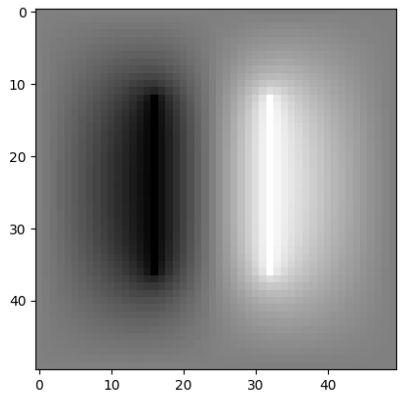

Figure 1: Converged electric potential using SOR ($\omega = 1.5$)

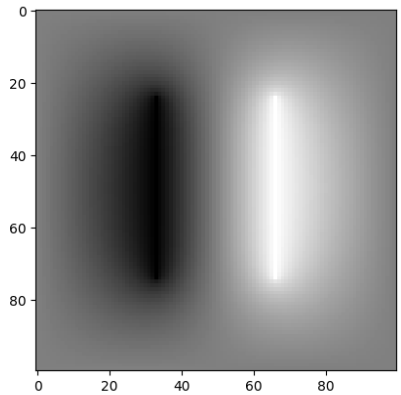

Figure 2: Converged electric potential using SOR ($\omega = 1.5$)

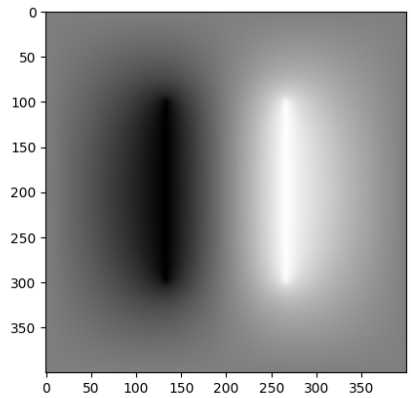

Figure 3: Converged electric potential using SOR ($\omega = 1.5$) for $N = 400$. The solution is visually identical across all grid sizes, confirming convergence to the correct potential distribution.

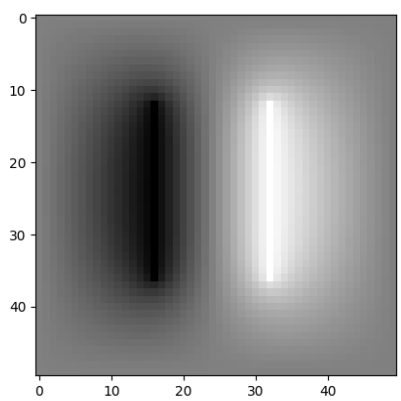

Figure 4: Converged electric potential using the Jacobi method for $N = 50$.

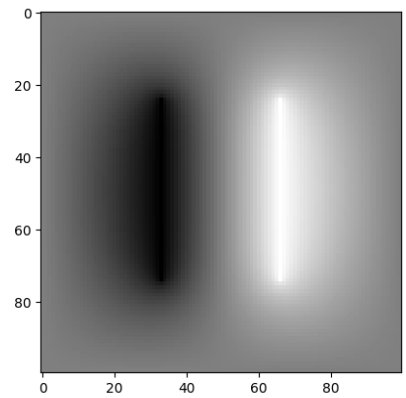

Figure 5: Converged electric potential using the Jacobi method for $N = 100$.

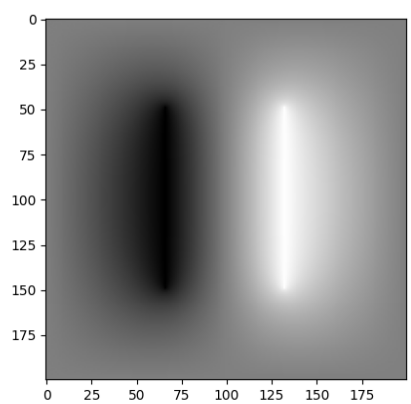

Figure 6: Converged electric potential using the Jacobi method for $N = 200$.

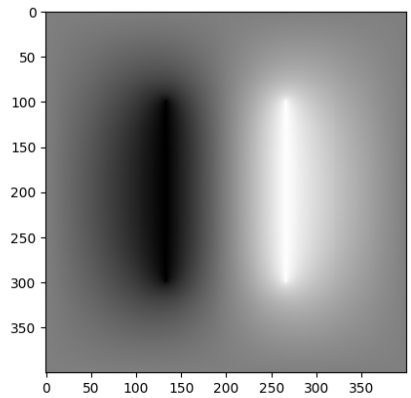

Figure 7: Converged electric potential using the Jacobi method for $N = 400$. Both the Jacobi and SOR methods converge to the same solution; they differ only in the number of iterations required.

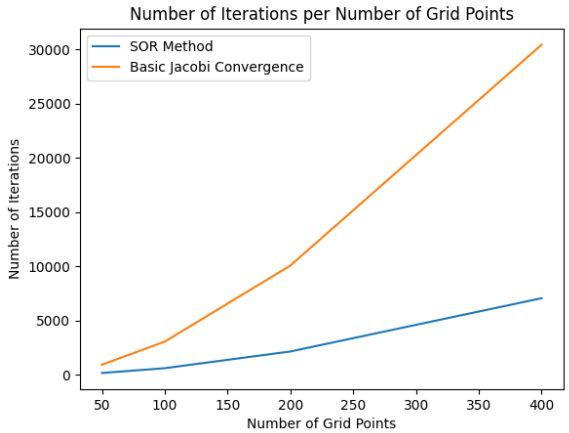

Figure 8: Number of iterations to convergence versus grid size N for the Jacobi method and SOR ($\omega = 1.5$). The Jacobi curve grows roughly quadratically with N while the SOR curve grows more slowly, demonstrating the advantage of over-relaxation at larger grid sizes.

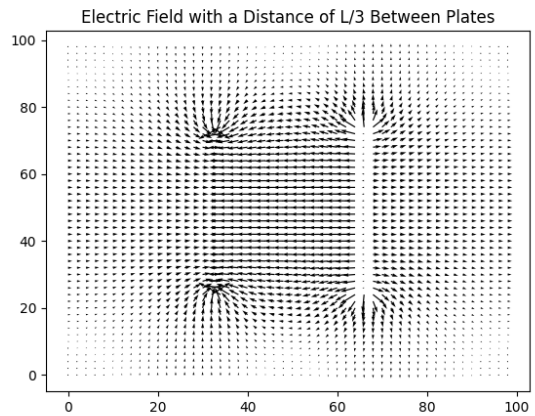

Figure 9: Electric field quiver plot for plate separation $d = L/3$. The field between the plates is nearly uniform and directed from the positive to the negative plate. Fringe fields fan outward at the plate tips.

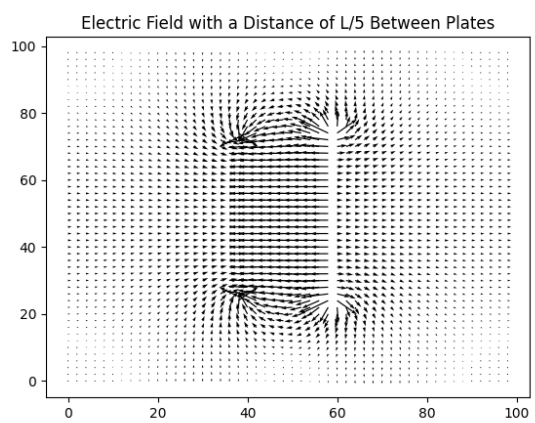

Figure 10: Electric field for plate separation $d = L/5$. The inter-plate field is stronger and more confined than in Figure **??**.

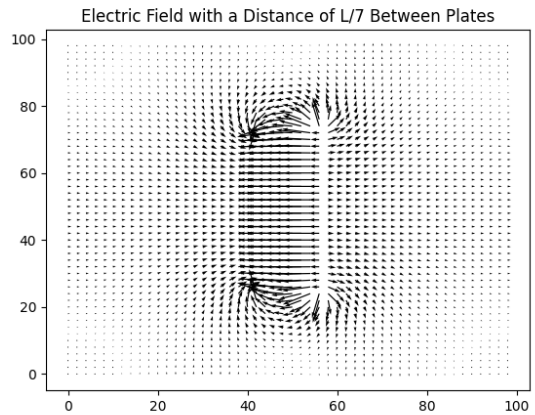

Figure 11: Electric field for plate separation $d = L/7$.

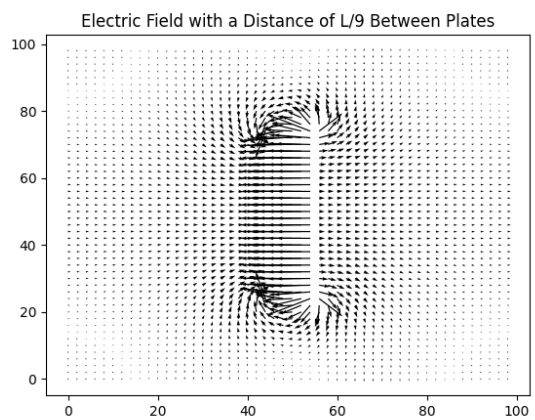

Figure 12: Electric field for plate separation $d = L/9$.

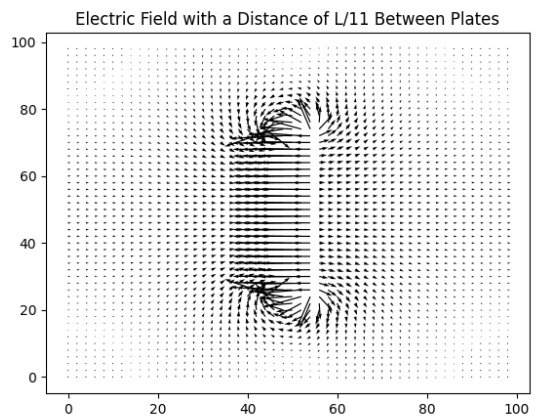

Figure 13: Electric field for plate separation $d = L/11$. Fringe fields are almost entirely confined to the plate tips; the inter-plate field is nearly indistinguishable from a uniform field, consistent with the parallel-plate approximation $|E| \approx \Delta V/d$.